Evaluation of background exposures of Americans to dioxin-like compounds in the 1990s and the 2000s.

The US Environmental Protection Agency's 2004 Dioxin Reassessment included a characterization of background exposures to dioxin-like compounds, including an estimate of an average background intake dose and an average background body burden. These quantities were derived from data generated in the mid-1990s. Studies conducted in the 2000s were gathered in an attempt to update the estimates generated by the Reassessment. While these studies suggest declines in the average background dose and body burden, a precise quantification of this decline, much less a conclusion that a decline has indeed occurred, cannot be made because of the inconsistency of study design and data sources, and the treatment of non-detects in the generation of congener average concentrations. The average background intake of the Reassessment was 61.0 pg TEQ/day, and using more current data, the average background intake was 40.6 pg TEQ/day. The average body burden from the surveys in the mid-1990s was 22.9 pg TEQ/g lipid weight (pg/g lwt). More recent blood concentration data, from NHANES 2001/2, suggest an adult average at 21.7 pg/g TEQ lwt. These TEQ values include the 17 dioxin and furan congeners and 3 coplanar PCBs, and were generated substituting ND=(1/2)DL or ND=DL/sq rt (2). Results are provided for ND=0 and analyses conducted to evaluate the impacts of this substitution. A more detailed examination of beef and pork data from similarly designed national statistical surveys show that declines in pork are statistically significant while the beef concentrations appeared to have remained constant between the time periods.